

Questions: Introduction to fractions

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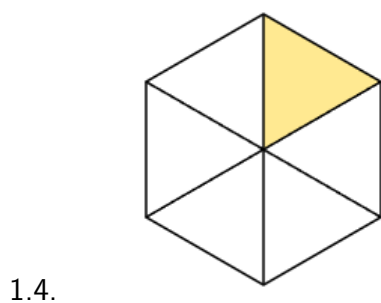
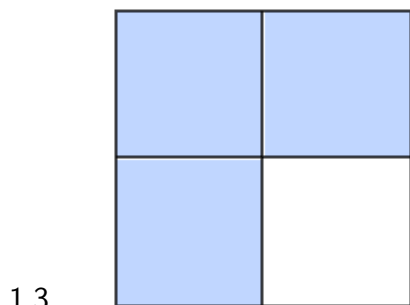
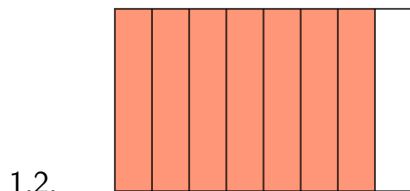
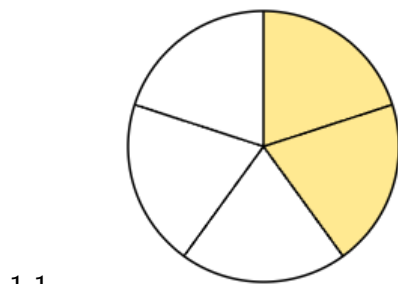
Summary

A selection of questions for the study guide on the introduction to fractions.

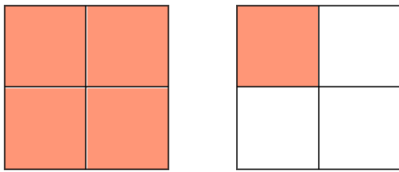
Before attempting these questions, it is highly recommended that you read [Guide: Introduction to numerical fractions](#). You may also find [Calculator: Highest common factor, lowest common multiple](#) useful.

Q1

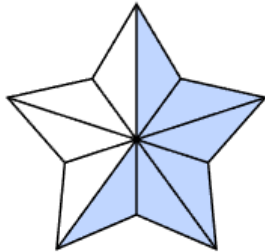
For each figure, write the fraction that represents the shaded area.



1.5.



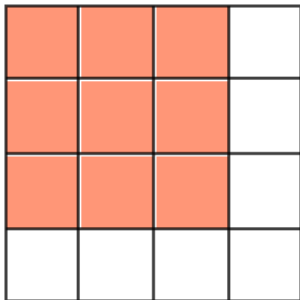
1.6.



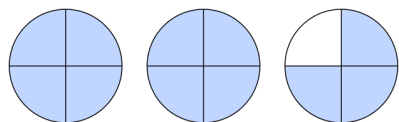
1.7.



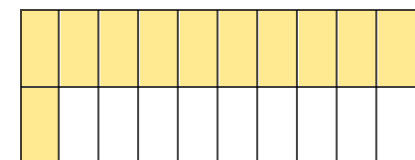
1.8.



1.9.



1.10.



Q2

Convert each mixed number into an improper fraction.

2.1. $1\frac{1}{2}$

2.2. $-2\frac{2}{3}$

2.3. $3\frac{1}{4}$

2.4. $-5\frac{2}{5}$

$$2.5. \quad 4\frac{3}{7}$$

$$2.6. \quad 6\frac{1}{6}$$

$$2.7. \quad -8\frac{3}{5}$$

$$2.8. \quad 10\frac{2}{9}$$

$$2.9. \quad -7\frac{5}{11}$$

$$2.10. \quad 12\frac{3}{4}$$

Q3

Convert each improper fraction into a mixed number.

$$3.1. \quad \frac{5}{2}$$

$$3.2. \quad -\frac{7}{4}$$

$$3.3. \quad \frac{10}{3}$$

$$3.4. \quad \frac{-11}{7}$$

$$3.5. \quad \frac{12}{6}$$

$$3.6. \quad -\frac{25}{6}$$

$$3.7. \quad \frac{31}{9}$$

$$3.8. \quad \frac{50}{11}$$

$$3.9. \quad \frac{64}{-8}$$

$$3.10. \quad -\frac{100}{13}$$

Q4

Find the missing value ? that makes the two fractions equivalent.

$$4.1. \quad \frac{1}{4} = \frac{?}{12}$$

$$4.2. \quad \frac{2}{3} = \frac{6}{?}$$

$$4.3. \quad \frac{3}{5} = \frac{?}{25}$$

$$4.4. \quad \frac{5}{8} = \frac{?}{16}$$

$$4.5. \quad \frac{3}{4} = \frac{?}{-20}$$

$$4.6. \quad -\frac{1}{6} = \frac{?}{24}$$

$$4.7. \quad \frac{5}{8} = \frac{20}{?}$$

$$4.8. \quad \frac{4}{3} = \frac{?}{18}$$

$$4.9. \quad -\frac{1}{3} = \frac{?}{27}$$

$$4.10. \quad \frac{7}{10} = \frac{70}{?}$$

$$4.11. \quad -\frac{4}{5} = \frac{?}{30}$$

$$4.12. \quad \frac{11}{12} = \frac{?}{60}$$

$$4.13. \quad \frac{3}{-7} = \frac{?}{21}$$

$$4.14. \quad \frac{8}{9} = \frac{32}{?}$$

$$4.15. \quad \frac{6}{7} = -\frac{?}{-49}$$

Q5

Write each fraction in its simplest form.

$$5.1. \quad \frac{4}{8}$$

$$5.2. \quad \frac{3}{9}$$

$$5.3. \quad \frac{6}{10}$$

$$5.4. \quad \frac{9}{12}$$

$$5.5. \quad \frac{15}{25}$$

$$5.6. \quad \frac{7}{21}$$

$$5.7. \quad \frac{20}{30}$$

$$5.8. \quad \frac{35}{49}$$

$$5.9. \quad \frac{48}{72}$$

$$5.10. \quad \frac{100}{120}$$

Q6

Convert each fraction into its alternative form and fully simplify the result.

- If an improper fraction is given, convert it into a mixed number.
- If a mixed number is given, convert it into an improper fraction.

$$6.1. \quad \frac{6}{4}$$

$$6.2. \quad 2\frac{2}{8}$$

$$6.3. \quad \frac{12}{10}$$

$$6.4. \quad -\frac{15}{9}$$

$$6.5. \quad 3\frac{4}{6}$$

$$6.6. \quad -1\frac{6}{8}$$

$$6.7. \quad \frac{20}{12}$$

$$6.8. \quad \frac{30}{25}$$

$$6.9. \quad 5\frac{10}{15}$$

$$6.10. \quad -\frac{45}{20}$$

$$6.11. \quad 4\frac{8}{10}$$

$$6.12. \quad \frac{50}{30}$$

$$6.13. \quad \frac{75}{-50}$$

6.14. $6\frac{12}{16}$

6.15. $-2\frac{14}{21}$

[After attempting the questions above, please click this link to find the answers.](#)

Version history and licensing

v1.0: initial version created 12/25 by Donald Campbell as part of a University of St Andrews VIP project.

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